

KAGAN ONUR

Data Analyst • Data Scientist

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PROFILE

Statistics graduate of Yıldız Technical University (2026). What interests me is getting from data to a decision: my thesis modelled 21 years of Turkish investment instruments to measure which one actually protects savings against inflation and currency risk; a separate project tested which patterns in twenty-four years of Turkish macro data genuinely repeat, and published the results; and I represented Türkiye at ASA Data Quest 2025 alongside teams from four continents. I work in Python, R, Stata and SQL, and I treat explaining *why* a model behaved as it did as part of the job rather than an afterthought. Alongside this I designed, built and shipped mesite.org single-handedly. Open to data analyst and data scientist roles.

SELECTED PROJECTS

The Turkish Economy — Finding the Rules That Repeat

Individual project • kaganonur.vercel.app/turkiye-ekonomisi • 2026, ongoing

Is there anything about the Turkish economy I can state as a general rule? I collected 33 monthly and daily series from 2002 onward and tested the usual claims one by one, writing up the ones that survived and the ones that collapsed. The output is five published sections and an interactive explorer covering 36 series.

- Seven rules survived. The two clearest: in **twenty-four years out of twenty-four** the current account is better in August than in March of the same year, and **82% of that swing** is explained by net tourism revenue; the real budget deficit peaks in December in fifteen of twenty years.
- Four claims failed — including "the lira gets volatile after elections" and "the budget opens up before elections." Every finding went through calendar-matched placebo and permutation tests, and the seasonality method itself through a negative control that my first version failed.
- **I retracted three of my own results:** the level Taylor coefficients were a spurious regression (DW 0.13–0.47, no cointegration), the Bai–Perron break dates vanished once residual autocorrelation was removed, and one relationship rested on a single observation. All three are written into the published pages.

Methods: Bai–Perron multiple structural breaks (my own dynamic programming implementation) • Chen–Liu outlier detection • calendar-matched placebo and permutation tests • Benjamini–Hochberg FDR • 2×12 seasonal decomposition with a negative control • lag-augmented local projections (Montiel Olea & Plagborg-Møller, 2021) • Zivot–Andrews • spurious-regression diagnostics • event study around rate-decision dates. Python (pandas, statsmodels), CBRT EVDS API.

Time Series Analysis of Turkish Investment Instruments

Undergraduate Thesis • Yıldız Technical University • 2026

Advisor: Prof. Dr. Öyküm Esra Yiğit

Which asset actually protects savings against Türkiye's inflation and currency swings — and can deep learning forecast these markets better than a classical model? I tested both questions on 21 years of monthly data across five series.

- Gold was the only instrument that held its real value over two decades; deposit returns were eroded by inflation.
- Adding the 2008, 2018 and 2021 crises to the model improved the gold forecast fourfold (R^2 0.12 → 0.51).
- Classical ARIMA beat the deep learning LSTM. I looked into why rather than taking the win: the comparison wasn't fair, because the LSTM forecasts on top of its own forecasts and the error compounds over 46 months.
- Confirmed that currency moves drive inflation rather than the reverse — a measure of exchange-rate pass-through.

Methods: ADF / PP / KPSS unit-root tests • Bai–Perron structural breaks • ARDL bounds testing • ARMA-GARCH(1,1) and EGARCH • Granger causality • ARIMA vs LSTM vs Holt benchmarking. Python (statsmodels, arch, TensorFlow/Keras).

MeSite — Branching Video Platform

Individual project • mesite.org • 2026, ongoing

A video platform where stories branch: you watch a clip, and if you don't like where it goes you upload your own continuation. I built the whole thing alone — data model, algorithm, API, interface and server — and it is live in closed beta.

- Designed the algorithm that decides which branch becomes the main storyline. I measured attention in **capped seconds rather than completion rate**, because completion rate would push everyone to cut their videos shorter.
- Made diversity and attention **multipliers rather than added points**, so a branch inflated by one author or watched by nobody can't become the main line however large it grows.

- Shipped it end to end: nightly backups with a tested restore, rate limiting, and account deletion under Turkish data protection law.

Stack: Python (FastAPI, SQLAlchemy) · PostgreSQL for accounts, Neo4j for the branching graph · Docker · Caddy · JWT · FFmpeg transcoding · JS/HTML/CSS front end. ~8,600 lines of Python with an automated test suite, deployed on a Hetzner VPS.

Effect of Price Discounts on Player Counts on Steam

Panel Data Analysis · Team of 3 · 2026

Do Steam sales actually bring players back? We tested it on twelve months of data from nine popular games.

- Price matters: a **1% increase costs 0.57% of the player base**.
- The sales themselves did not. None of the four seasonal sale periods produced a significant lift.
- The reason was a measurement trap: **98% of the difference in player numbers comes from how big the game already is**. Discounts don't grow a game — large games simply stay crowded.

Methods: balanced panel of 108 observations (9 games × 12 months) · a combined "discount pressure" variable merging depth and duration · panel unit-root tests · Hausman test selecting fixed effects · game-clustered robust standard errors. Stata, R.

Data Science & Sustainable Development

ASA Data Quest 2025 · International · Representing Türkiye

American Statistical Association (CIRS) · Mentor: Ceylan Yozgatligil · Team of 4

Do countries have to accept environmental cost in order to develop? We presented Türkiye's answer against teams from Brazil, Indonesia, India, Kenya and China.

- Wealth, education and life expectancy rise together almost in lockstep (**0.87–0.92**).
- Carbon cost is where they part: national income tracks emissions closely (**0.87**) while education and health spending do not — social investment develops a country at a lower carbon price.

Validation: Durbin–Watson, VIF and Shapiro–Wilk diagnostics.

EXPERIENCE

Data Annotation Specialist

Jan 2026 – Feb 2026

Artiwise — AI-Powered Software Company

- Labelled and classified text data for supervised NLP models at a company building AI-powered document analysis for customer-centric process management.
- Applied and refined annotation guidelines to keep labels consistent, and ran quality checks on completed batches to keep training data reliable.

EDUCATION

Yıldız Technical University

2022 – 2026

BSc Statistics, Faculty of Arts and Sciences

Relevant coursework: Econometrics · Time Series · Panel Data · Multivariate Statistics · Regression · Data Visualization

İstanbul Kültür University

2020 – 2022

Computer Engineering, Faculty of Engineering

English preparatory programme completed; transferred to Statistics at YTU

Gürlek Nakipoğlu Anatolian High School

2015 – 2019

TECHNICAL SKILLS

Programming & Tools	Python (pandas, NumPy, statsmodels, scikit-learn, TensorFlow/Keras) · R · Stata · SQL · Excel · SPSS
Time Series & Econometrics	ARIMA / SARIMA · ARMA-GARCH · EGARCH · ARDL bounds testing · Granger causality · Bai–Perron · local projections · Zivot–Andrews · ADF / PP / KPSS
Panel & Regression	Fixed and random effects · Hausman test · cluster-robust standard errors · panel unit roots · OLS · logistic regression
Inference & Diagnostics	Placebo and permutation tests · multiple-comparison correction (Benjamini–Hochberg) · event studies · spurious regression and assumption audits
Databases & Engineering	PostgreSQL · Neo4j (graph database) · FastAPI · Docker · Git · REST API design
Machine Learning	LSTM and deep sequence models · supervised classification · model benchmarking (MAE, MAPE, R ²)
Data Work	Cleaning and merging multi-source data · feature engineering · data labelling · visualization and reporting
Languages	Turkish — Native · English — Upper-Intermediate (B2), DİLKO in partnership with Pearson (2022)